Design and Development of Cyber Physical System for Real-Time Web-based Visualization and Control of Robot Arm

Suwanan Khrueangsakun, Santi Nuratch, Panuthat Boonpramuk
Embedded Computing and Control Laboratory
Department of Control System and Instrumentation Engineering, Faculty of Engineering, KMUTT, Bangkok, Thailand
e-mail: suwanan.k@mail.kmutt.ac.th, santi.inc.kmutt@gmail.com, panuthat.boo@kumtt.ac.th

***Abstract*— Since Cyber-Physical System or CPS currently takes part in industrial, there is a lot of research for developing this system. From the survey and the trend of CPS research, a human user interface for control and monitor was highlighted as Digital Twins where VR and AR are introduced to improve the visualization for easier system interpretation. This paper introduces the proposed design of real-time control web-based monitor and control of 4-DoF robotic arm system where the robotic arm represents a physical part and web-based visualization as a cyber-part. The 4-DoF robotic arm was utilized from MeArm robot and was controlled via ESP32 WROOM with Wi-Fi module in order to pursue wireless data transmission. WebSocket protocol will be exploited to connect between the physical and cyber-part where the virtual model was displayed and controlled via web-browser, the virtual model of the robot was created with 3D blender and operated with WebGL-based 3D application which can be integrated and run on web-browser without extension. The proposed design was evaluated with operational time and error between the controlled virtual model and actual physical angle, the result achieved 278.772 ms for average operational time and 0.32% in error of angle between virtual and actual. As a result, the proposed design provided two-way communication, real-time control, and monitor which can be applied in industrial applications.**

Keywords—CPS, real-time monitoring and control, robot arm, virtual simulation, web-based application

I. INTRODUCTION

Cyber-Physical System (CPS) is the system which combines the machine in industrial or process as a physical world and high-level human interface or computational system as a cyber world. Both worlds are connected wirelessly with internet communication, thus, users can retrieve data remotely and utilize the data in order to control or analyze the system. Consequently, CPS can be integrated with cloud technology which can be applied to industrial improvement [1]. Since CPS delivers the accuracy, efficiency, data-centric monitoring platform and real-time control, there are several studies related to industrial control and manufacturing. From the review of CPS survey for control engineering [2], IoT connection was presented, the research mentioned on the human-machine interface (HMI) that was responsible for monitor and control in CPS, the main goal of the research focused on improvement that will provide more user-friendly to manage the system. Moreover, virtual modeling and simulation were pointed out in the concept of Digital Twins. The study focuses on the behavior of the interaction between two worlds which will provide an advantage for the process planning and monitoring the different scenarios.

As HMI and digital twins are accountable for an interface between users and machines, the visualization is important. The earlier research presented the non-virtual reality visualization for monitoring and control [3], yet, there was a limitation in terms of convenience and accessibility. By pressing a button or inject the angle to control and ambiguous image or model of an object like a robot arm, users require time to understand the whole view and need to know the parameter before controlling which isn't user-friendly and inconvenient. On one hand, AR and VR were being increasingly applied in industrial [4], with AR and VR Modeling, engineers or even general people can understand the whole system easier with image or model, save time and prevent the damage on the physical robot in trajectory planning task [5]. Besides, both technologies can be applied for training and maintenance guidance applications for the manufacturing line [6],[7].

Meanwhile, the study was considerably highlighted on robot control, the most related research similar to the concepts of Digital Twins was Virtual Reality Model for 5-DoF robot arm control with Simulink model in MATLAB [8]. Users can control robot simulation and result in the physical robot to move according to the simulation, however, users need an additional program installation like MATLAB in order to monitor and control. This system can be developed by employing two-way communication. In other words, if there's an incident on the physical side, the simulation will acknowledge and display the real-time monitor. Furthermore, to deliver convenience for users, the application should be easily accessible without additional programs.

Consequently, this research aims to design and develop the CPS system which consists of the robotic arm as the physical side, 3D model simulation on the cyber side for real-time monitoring and control on web-application where both sides were linked by WebSocket. The paper is organized as followings, Section II introduces the system overview of the proposed design. Section III demonstrates the experiment set up and includes the result and discussion respectively. Finally, the conclusion and future work will be described in Section IV.

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II. Proposed System Design

A. System Overview

CPS consists of three parts; physical, cyber and connection between both worlds, the 4-DoF robot arm was presented as physical part while the 3D simulation was operated with real-time 3D realistic WebGL based in order to display and control the robot arm. Both physical and cyber-part communicate with each other via WebSocket. The robotic arm will be manipulated with ESP32 (Wi-Fi module included) for control servo motors and transfer data between robot and 3D virtual model.

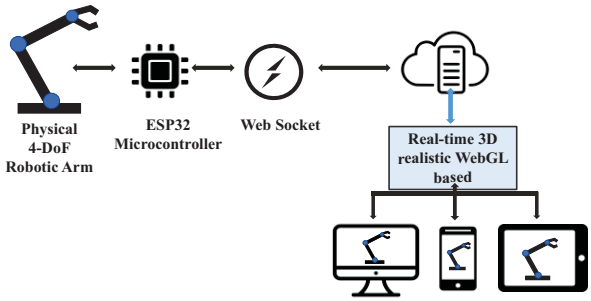

Figure 1. System Overview of Proposed Design.

B. Hardware: 4-DoF Robotic Arm and ESP32

The hardware side as a physical part introduces with 4-DoF robotic arm with four servo motors which will be driven with ESP32-WROOM-32 Module. Furthermore, the motor can be controlled real-time and wireless with Wi-Fi Module within ESP32 MCU.

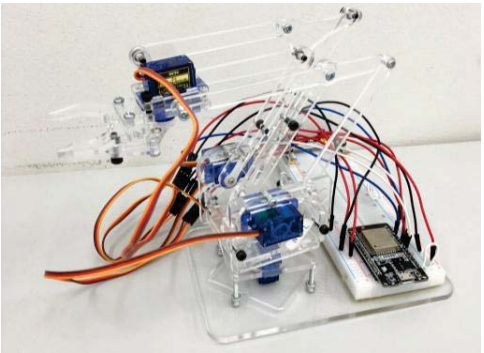

Figure 2. Robot arm with ESP32 MCU.

C. WebGL

The real-time 3D realistic WebGL-based 3D application [9] consists of two main parts; Engine Core and Engine API (Application Programming Interfaces) where both engines are integrated on website, then operate on the target web-browser. Engine API involves the necessary functions, for instance, asynchronous download the 3D model operation and others. At the same time, Engine core was integrated with the THREE graphic library in order to display the 3D virtual model. With this engine, model or any 3D visualization can be imported, rendered and operated control event directly to web-browser faster and easier without any extension.

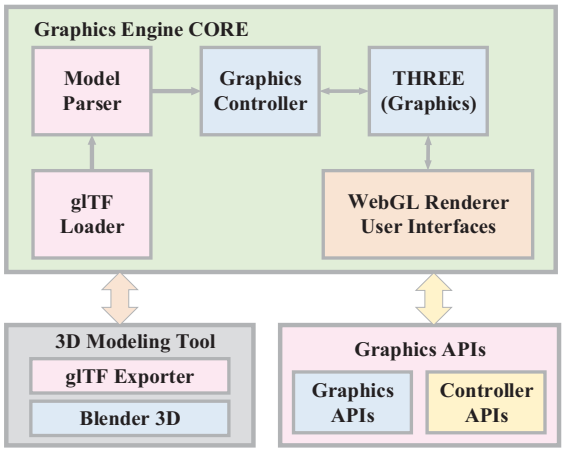

Figure 3. WebGL-base 3D Application Engine Architecture.

For virtual simulation model of robot arm on web-browser, the model was generated with Blender and handled with WebGL on web-browser as shown in Fig. 4.

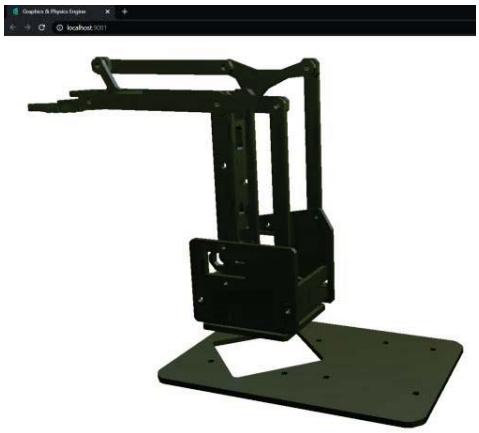

Figure 4. Virtual simulation of robot arm running with WebGL engine.

D. WebSocket

WebSocket is a full-duplex communication protocol that gives the two-way communication and serves real-time data transmission between client and server. This protocol takes only the first time of HTTP requested to establish the TCP socket connection, subsequent with WebSocket reader protocol [10]. Due to its advantages; real-time, less data loss and double open communication, it satisfies the proposed design where the data should be transfered real-time in both control and monitoring, where, the ESP32 and 3D virtual model web-app serve as the client.

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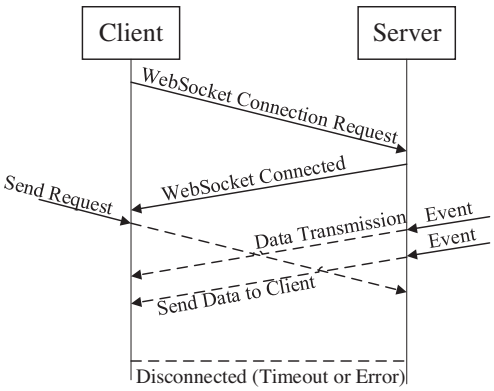

Figure 5. WebSocket architecture.

III. EXPERIMENTAL SETUP AND RESULTS

To evaluate the system, operational time and the error between the desired angle from the virtual simulation of the robot arm on the web-browser and actual angle of physical robot arm will be inspected.

As the proposed system focus on real-time data transfer, the operational time was evaluated with the local server and remote server in order to ensure the remote control. With a foreign country server, web-browser with virtual model represent the cyber-part and client of the WebSocket communication, the server simulated to operate in Singapore. The limitation of the physical angle as shown in Table I.

TABLE I. LIMITAION OF ROBOT ARM MOVEMENT ANGLE

Joint No.	*Min*	*Max*	*Joint No.*	*Min*	*Max*
Joint 1	0	180	Joint 3	30	180
Joint 2	90	180	Joint 4	90	180

Operational time was measured when the data as an angle to control the robot was transferred from web-browser to microcontroller. After the user finished the angle adjustment of the virtual robot, the microcontroller will respond when the physical robot's actual angle achieved the desired angle back to the web-browser. The total time of operation is calculated as the following equation.

$$t_{op} = t_{mcu} + 2t_{trans} + 2t_{lat} + t_{web} \quad (1)$$

t_{op} presents the total operation time, t_{mcu} is microcontroller computational time, t_{trans} denotes data transmission time, t_{lat} stands for latency of network and t_{web} presents the computational time for web-browser on the 3D virtual simulation web.

TABLE II. OPERATIONAL TIME

Operation Time (Server)	*Time (milliseconds)*		
	min	**max**	**avg**
Joint Control (Local)	2	1183	288.7772
Joint Control (Foreign)	502	1704	823.123

Refer to Table II, the computational time is obviously different between local and foreign servers. The local server provided faster operational time which is responsible for 288.7772 ms, this yields the acceptable time to control and monitor in real-time which presents the strength of this work. While the foreign server gives average at 823.123 ms, this happens when users desire to control the system from other countries, the data transmission has to pass multiple gateways and nodes before sending data to the local system, consequently, the operational time extends longer.

On the other hand, when users abruptly control the angle from virtual simulation, the physical robot arm doesn't respond immediately with the simultaneously changing virtual simulation angle. To demonstrate, the user adjusts the angle from 90 to 120 degrees, while the physical robot is moving to the desired angle (120 degrees), the user suddenly controls to the new desired angle (60 degrees). The robot takes more time to get itself to the new desired angle. In addition, the microcontroller sometimes ignores some angle between the ranges from the previous angle and new adjustment. Consequently, sampling time was included in order to feed the data in a period and allow the microcontroller to operate functionally. The results were given as the following table.

TABLE III. OPERATIONAL TIME WITH LOCAL SERVER

Operational Time (Sample Time)	*Time (milliseconds)*		
	min	**max**	**avg**
Joint Control (50 ms)	13	701	241.442
Joint Control (100 ms)	40	2476	625.933
Joint Control (500 ms)	64	5418	739.145

TABLE IV. OPERATIONAL TIME WITH FOREIGN SERVER

Operational Time (Sample Time)	*Time (milliseconds)*		
	min	**max**	**avg**
Joint Control (50 ms)	513	1024	741.124
Joint Control (100 ms)	643	3017	1243.76
Joint Control (500 ms)	701	6079	1235.68

According to Table III and IV, adding 50 ms offers the fastest operational time, namely, 241.442 ms for the local server, in the same way, the foreign server gives 741.124 ms. By adding the sampling rate, the data was steadily fetched to the microcontroller, it prevents data loss from abrupt control, if users want to send every single controlled angle to the robot arm system without any loss, it requires more time to operate. Therefore, it's a drawback to choose between speed and correctness with more time to process. Furthermore, with more sampling time, it tends to lessen the sudden jiggle when users suddenly control the virtual angle of the robot. This can be improved in the future by applying the sigmoid function and switching control to gradually feed the angle data to the hardware.

Next part of the experimental setup, the comparison between the actual angle of the physical robot arm and the virtual model on the web browser that will be controlled and simulated. The actual angle of the physical robot will be measured with encoder. The graph of error was shown as

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following (Fig. 6), where the Y-axis stands for the error between controlled virtual angle from users and the actual angle of the robot arm.

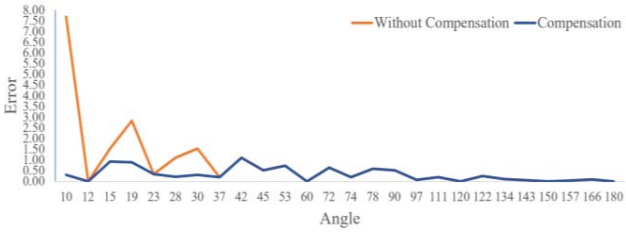

Figure 6. Error between controlled and actual angle.

According to the result, the error occurred when the controlled angle was small between 10 to 20 degrees. The reason is the pulse to drive the motor wasn't enough and need to be compensated. After applying 5% of pulses for a small angle, the error eased from 0.78% (without compensation) to 0.32% (with compensation). On the cyber-part, WebGL was implemented in web-browser to display the 3D virtual robot arm, which can be operated on a web-browser with 59-61 frames/second. The virtual model will change corresponding to the physical robot arm as the following figure. Besides, this error might occur in a specific system using this robot arm and microcontroller.

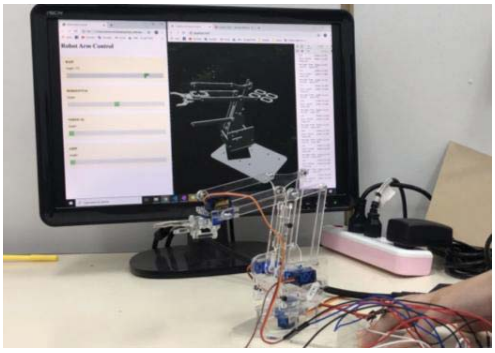

Figure 7. The virtual model and physical robot arm.

For the 3D visualization, it allows users to understand the whole system easily and control directly with the 3D model without knowing the parameter of the robot joint. This can be contributed to implementing in the industry for example grasp and grip task or complex machine assemble tasks that need to be achieved by a human. Besides, the proposed system can be developed to contribute to another part that requires the illustration or simulation to interpret the system in the industry like path and trajectory planning, maintenance, etc.

IV. Conclusion and Future Work

After all, this paper introduces the system where the CPS concept was adapted. The robot arm represents the physical part of the system which can be controlled wireless through the web-browser, communication was carried out with the WebSocket protocol which provides 288.7772 ms to connect and respond to control and visualize the virtual simulation model. For the cyber-part, the 3D virtual simulation model was displayed for the user, this can be controlled in real-time and directly change to the physical arm where the error of the performance after pulse compensation namely 0.32%. The error can be guaranteed the result of the virtual and actual angle to develop when applying the kinematic to the robot arm in the future. This proposed system could be contributed to apply in the industrial for control since it provides the fast response which is an advantage for real-time control in manufacturing. For future work, the improvement will be focused on the less jiggle movement of the robotic arm and more real-time response by code optimization. Moreover, the kinematic system will be implemented to the model to afford more realistic movement of robot arm.

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